

MediLink Healthcare Appointment Platform

Agile Delivery Plan — 8-Week Sprint Cycle with Weekly Demos

Document	Agile Delivery Plan (Final)
Date	2026-06-16
Prepared By	Satyam Kumar
Delivery Cadence	8 weeks • 4 sprints • 2 weeks per sprint • weekly demos
Products	Patient Mobile App (Flutter) & Patient Web App (Next.js 15)
Status	For stakeholder review & approval

Alignment: This plan delivers the MediLink patient platform over an 8-week Agile cycle with weekly demonstrations, in line with the agreed development approach. All capabilities are organised in a single unified roadmap, prioritised so that the revenue-generating booking journey is delivered first and intelligent (AI) features build on a stable core.

1. Executive Summary

MediLink is a bilingual (English / Arabic RTL) healthcare appointment platform delivered as two patient-facing products — a Flutter mobile app and a Next.js 15 web app — built on a single Supabase backend and deployed on Vercel + Supabase. This plan delivers the complete patient experience over an 8-week Agile cycle of four 2-week sprints, with a working demonstration at the end of every week.

- Outcome:** patients can register, discover doctors, book and manage appointments, pay securely via Thawani, receive invoices and notifications, and access richer features (family members, records, reviews) and AI-assisted discovery — all bilingual.
- Prioritisation:** the booking-and-payment journey is delivered first; engagement, records and AI features follow within the same roadmap.
- Transparency:** a deployable increment is demonstrated weekly, giving stakeholders continuous visibility and early opportunity to steer.
- Quality:** overbooking protection, payment reconciliation and full EN/AR coverage are built in and verified, not deferred.

2. Project Scope

In scope — Patient Mobile App (Flutter) and Patient Web App (Next.js 15):

Area	Capabilities
Authentication	Email login, OTP verification, Google login, forgot password
Patient Profile	Profile management, medical history
Doctor Discovery	Search doctors, filters, doctor profile, facility profile
Appointments	Book, reschedule, cancel, slot locking, buffer time, overbooking protection, web check-in
Payments	Thawani integration, payment history, invoice generation
Notifications	Email, push, in-app notifications
Localization	English, Arabic (RTL)
Additional Patient Features	Family members, document vault, insurance details, ratings & reviews, follow-up booking, multi-location booking

Area	Capabilities
AI Features	AI symptom checker, AI doctor suggestions, AI scheduling assistant

Out of scope: This engagement covers the patient products only. Any provider, clinic, doctor, technician, administrative or facility-side applications are not part of this delivery.

3. Assumptions

- Doctor, facility and availability data are provided to the platform (seeded and maintained externally); the patient apps consume this data and do not manage it.
- Thawani merchant credentials and sandbox/production access are available at the start of the payment sprint.
- Google OAuth client credentials and push-notification provider keys (FCM / APNs) are available when needed.
- Final UI/UX direction (branding, colours, typography) is confirmed before or during Sprint 1.
- English and Arabic content/translations are supplied or signed off as screens are built.
- Weekly demonstrations are reviewed promptly so feedback can be incorporated into the next increment.
- Scope is fixed for the 8-week cycle; new requirements are logged and scheduled after this delivery.

4. Technology Stack

Layer	Technology
Mobile (Frontend)	Flutter (Android & iOS)
Web (Frontend)	Next.js 15 (App Router)
Backend	Next.js Route Handlers + Server Actions (where appropriate)
Database	Supabase PostgreSQL (Row-Level Security)
Authentication	Supabase Auth
Storage	Supabase Storage
Realtime	Supabase Realtime
Deployment	Vercel + Supabase
Languages	English + Arabic (RTL)

Both the mobile and web apps consume a single, shared backend built with Next.js Route Handlers and Server Actions over Supabase, ensuring consistent business logic and a single source of truth across platforms.

5. 8-Week Delivery Timeline

Week	Sprint	Focus	Weekly Demo Milestone
1	Sprint 1	Foundation, schema, environments; authentication	Apps deployed to staging; login flow taking shape
2	Sprint 1	Auth complete + patient profile & medical history	Register, OTP, Google login + profile (EN/AR)
3	Sprint 2	Doctor discovery: search, filters, profiles	Search + view doctor / facility profiles
4	Sprint 2	Booking, slot locking, buffer, overbooking, check-in	Book / reschedule / cancel + web check-in
5	Sprint 3	Thawani payments, invoices, payment history	Paid booking + downloadable invoice
6	Sprint 3	Notifications + family, vault, insurance, reviews	Notifications live; records & reviews
7	Sprint 4	Follow-up & multi-location booking + AI features	Follow-up rebooking; AI symptom checker

Week	Sprint	Focus	Weekly Demo Milestone
8	Sprint 4	AI assistants, Arabic RTL hardening, UAT, launch	Launch candidate; UAT sign-off

Each weekly demo presents a working, deployable increment. Capabilities are sequenced so the core booking-and-payment journey is usable early, with engagement and AI features layered on top.

Sprint Details

Each sprint below sets out its objective, representative user stories, acceptance criteria, risks and deliverables. Demonstrations are held at the end of each week.

Sprint 1 — Foundation, Authentication & Patient Profile

Duration: Weeks 1-2

Weekly demos: W1: staging deployment + auth in progress. W2: full auth + profile.

Objective: Establish the shared platform foundation and deliver secure authentication and patient profile management on both mobile and web, bilingual from day one.

User Stories

- bullet As a patient, can register and log in with email and verify my identity via OTP.
- bullet As a patient, can sign in with Google.
- bullet As a patient, can reset my password if I forget it.
- bullet As a patient, can create and edit my profile and record my basic medical history.
- bullet As a patient, can use the app in English or Arabic with correct right-to-left layout.

Acceptance Criteria

- bullet Apps build and deploy to staging automatically; database secured with row-level access.
- bullet Registration, OTP verification, Google login and password reset all work on web and mobile.
- bullet Profile and medical history save and persist across sessions.
- bullet Arabic renders with full RTL layout on every authentication and profile screen.
- bullet Provider/availability data is seeded and visible to support later booking.

Risks

- bullet Availability data depends on external seeding. **Mitigation:** agree the data contract and seed in Week 1.
- bullet OAuth redirect/deep-link differences between web and mobile. **Mitigation:** configure and device-test early.
- bullet OTP deliverability. **Mitigation:** verified email provider with fallback logging.

Deliverables

Deployed mobile + web foundation, secured database, and working authentication and patient profile in both languages.

Sprint 2 — Doctor Discovery & Appointment Booking

Duration: Weeks 3-4

Weekly demos: W3: discovery demo. W4: full booking lifecycle + check-in.

Objective: Enable patients to find the right doctor and manage the full appointment lifecycle, with robust protection against double-booking.

User Stories

- bullet As a patient, can search and filter doctors by specialty, location and availability.

- bullet As a patient, can view a doctor's profile and the facility's profile before booking.
- bullet As a patient, can book an appointment in an available slot.
- bullet As a patient, can reschedule or cancel an appointment.
- bullet As a patient, can check in for my appointment online and see its status.

Acceptance Criteria

- bullet Search returns correctly filtered results; web doctor/facility pages are server-rendered for SEO.
- bullet Two patients cannot book the same slot — exactly one booking succeeds under concurrency.
- bullet Buffer time is enforced between appointments; overbooking is prevented.
- bullet Reschedule frees the previous slot atomically; cancellation releases the slot.
- bullet Web check-in shows the correct, up-to-date appointment status.

Risks

- bullet Booking race conditions are the key technical risk. **Mitigation:** database-level slot locking + dedicated concurrency tests.
- bullet Search performance as data grows. **Mitigation:** indexed queries and pagination.

Deliverables

Complete doctor discovery and a fully working book / reschedule / cancel / web check-in journey on both platforms, with overbooking protection verified.

Sprint 3 — Payments, Invoices, Notifications & Patient Records

Duration: Weeks 5-6

Weekly demos: W5: paid booking + invoice. W6: notifications + records & reviews.

Objective: Close the revenue loop with Thawani payments and invoicing, deliver multi-channel notifications, and add core patient-records and engagement features.

User Stories

- bullet As a patient, can pay for my appointment securely through Thawani.
- bullet As a patient, can view my payment history and download invoices.
- bullet As a patient, receive email, push and in-app notifications for confirmations and reminders.
- bullet As a patient, can manage family members, store documents in a secure vault, and record insurance details.
- bullet As a patient, can read and leave ratings and reviews for doctors and facilities.

Acceptance Criteria

- bullet Payment confirms the appointment; failed or expired payment releases the held slot; reconciliation is reliable and idempotent.
- bullet Invoices are generated, downloadable, and listed in payment history; payment success rate $\geq 95\%$ in testing.
- bullet Confirmation and reminder notifications are delivered across email, push and in-app channels.
- bullet Family members, document vault and insurance details are saved securely with appropriate access controls.
- bullet Reviews can be added only after a completed appointment and display on the relevant profile.

Risks

- bullet Payment webhook reliability / idempotency. **Mitigation:** idempotency keys, signature verification, reconciliation job.
- bullet Protection of personal health information in storage. **Mitigation:** row-level security, signed URLs, private buckets.

Deliverables

Live Thawani payments, invoicing and payment history; multi-channel notifications; and family members, document vault, insurance and ratings & reviews on both platforms.

Sprint 4 — Advanced Booking, AI Features, Hardening & Launch

Duration: Weeks 7-8

Weekly demos: W7: follow-up/multi-location + AI symptom checker. W8: AI assistants, UAT, launch.

Objective: Add advanced booking options and AI-assisted experiences, complete bilingual hardening, run acceptance testing, and prepare the platform for launch.

User Stories

- bullet As a patient, can quickly rebook a follow-up appointment with the same doctor.
- bullet As a patient, can book across multiple branches of a facility.
- bullet As a patient, can use an AI symptom checker for guidance and be routed to relevant doctors.
- bullet As a patient, receive AI-based doctor suggestions and scheduling assistance when booking.
- bullet As a patient, experience a fully bilingual, polished app ready for real use.

Acceptance Criteria

- bullet Follow-up rebooking completes in a few taps; multi-location booking offers branch selection.
- bullet AI symptom checker returns guidance with a clear medical disclaimer and never provides a definitive diagnosis.
- bullet AI doctor suggestions and scheduling assistant return relevant, helpful results.
- bullet Full English/Arabic coverage verified across all screens; cross-device checks pass.
- bullet User acceptance testing is completed and signed off; launch checklist is green.

Risks

- bullet AI medical-safety and accuracy. **Mitigation:** guardrails, disclaimers, no diagnosis, route to a qualified doctor.
- bullet Late-cycle scope pressure. **Mitigation:** feature freeze entering Week 8; final week reserved for hardening, UAT and launch.

Deliverables

Follow-up and multi-location booking, AI symptom checker / doctor suggestions / scheduling assistant, full bilingual hardening, completed UAT, and a launch-ready patient platform on web and mobile.

10. Risks & Dependencies

#	Risk / Dependency	Impact	Likelihood	Mitigation
1	External provider / availability data required for booking	High	High	Agree data contract and seed in Week 1; apps consume, not manage
2	Booking race conditions / double-booking	High	Medium	Database-level slot locking + buffer rules + concurrency test suite
3	Thawani payment reliability & reconciliation	High	Medium	Idempotent webhooks, signature verification, reconciliation job
4	Google OAuth deep-link configuration (web vs mobile)	Medium	Medium	Configure URI schemes early; test on real devices
5	AI accuracy and medical-safety	High	Medium	Guardrails, disclaimers, no diagnosis, route to a doctor
6	8-week scope intensity	High	Medium	Strict prioritisation, weekly demos to surface slips early, Week-8 hardening freeze
7	Protection of personal health information	High	Low	Row-level security, signed storage URLs, private buckets
8	Translation / content readiness (EN/AR)	Medium	Medium	Supply or sign off content as screens are built

11. Success Metrics

- bullet **End-to-end booking:** a patient can complete register → discover → book → pay → confirm on both web and mobile.
- bullet **Booking efficiency:** the booking journey completes in under 90 seconds in usability testing.
- bullet **Payment reliability:** payment success rate ≥ 95%; zero double-charges (idempotent reconciliation).
- bullet **Booking integrity:** zero double-bookings under concurrency testing.
- bullet **Bilingual quality:** 100% of screens verified in English and Arabic (RTL).
- bullet **Notifications:** confirmation and reminder notifications delivered reliably across email, push and in-app.
- bullet **Delivery cadence:** a working, deployable increment demonstrated every week for all 8 weeks.
- bullet **Acceptance:** UAT completed with no open critical (Sev-1 / Sev-2) defects at launch.